

The Rank-Defect Theorem

The generalization beyond the ring: which stationary processes satisfy the equivalence of the arrow of time with a Hankel rank defect. Soundness holds — a visible complex mode is a sound arrow witness — but off the abelian slice both failure modes are generic: hidden arrows on real-spectrum chains (positive measure in the Birkhoff polytope) and fake arrows in the Gaussian stratum. On the N-ring the reversible slice is exactly the spectral-merging locus, the rank doubles with the arrow, and the reflection-laundering theorem shows the ring's arrow is its chirality: the ~~Amin Abae~~ arrow exactly when it loses the handedness. With the chat's Experiments 1–3 run at full audit strength: the AAK interval exact, the register explosion $9 \rightarrow 9153$, the sheaf ordering perfect at the task level and null at the seed level.

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1 · The mandate

Task 14 closed on two open frontiers, and the order that opens this volume names them exactly: the rank-defect theorem's generalization beyond the ring — the question of which stationary processes actually satisfy the equivalence arrow iff rank defect that the 3-state ring exhibited — and the chat's Experiments 1 through 3, the three legs of the empirical program that the DeepSeek thread had designed and this programme had not yet run. The first is a classification problem about the mathematical objects this corpus has carried since Volume II: the arrow of time as a reversal divergence, the Hankel rank as the memory stratum, and the sensor as the record's boundary. The second is an audit problem: three experiments, each with a published prediction, each to be run at the programme's full discipline — anchor-first, exact where exactness is available, honestly labelled where it is not.

The recovered chat defines the three experiments precisely (transcript lines 3497-3556). Experiment 1 hypothesizes that the Hankel singular values of a trained sequence model predict the optimal compression rate, that the AAK-optimal extraction beats pruning and distillation baselines at matched parameter count, and that the distortion-rate curve matches the σ_{n+1} lower bounds. Experiment 2 hypothesizes that the obstruction datum of a partially observed control problem predicts when a finite-state policy fails to exist and that the viability-weighted curvature measures the severity. Experiment 3 hypothesizes that the sheaf cohomology of a network's local computations predicts training failures, with the harmonic norm correlating with the validation gap. Each of these is tested below on the branch of its own benchmark family that admits exact machinery, and each returns a split verdict that this volume records without varnish.

The ordered link	The deliverable	Status	Verification
(1) The rank-defect theorem's generalization beyond the ring — which stationary processes satisfy $\text{arrow} \Leftrightarrow \text{rank defect}$	The classification: T1 soundness (reversible $\Rightarrow$ real spectrum, so a visible complex mode is a sound arrow witness); T2 the two failure modes with exact witnesses and the positive-measure scan; T3 the N-ring rank-doubling law (all N, the iff's natural home); T4/T4b the one-way valve (DPI) and the reflection-laundering theorem — the ring's arrow is its chirality	PROVED (T1, T3, T4, T4b) and WITNESSED (T2, both failure modes, both directions)	rank_defect.py: 1000/1000 reversible chains real-spectrum; the B3 hit-and-run 68.6% real-spectrum; the rational spectrum certificate exact; the rank law 6/6 for $N = 3..8$; the block KLs exact; the reflection theorem's kept/laundered split exact
(2) The chat's Experiments 1-3 — the spectral compression of sequence models, the sensor-coarsening scan, the sheaf diagnostics	Experiment 1 on the chat's own synthetic branch (the driven 6-ring through a 4-symbol sensor, a known-minimal WFA): the exact Gramian machinery, the AAK interval, the baseline triad. Experiment 2 on the chat's Tiger branch: the exact alpha-backup, the obstruction/register/curvature scan, the record-side exchangeability theorem. Experiment 3 on the compositional split: the token-sheaf coboundary energy vs the OOD gap	RUN at full audit strength; every one of the chat's quantitative predictions adjudicated honestly	exp1/exp2/exp3 batteries with results JSONs: the AAK interval 10/10; the register 9 -> 9153; the exchangeability deviation below $3.5e-18$; the task-level-perfect / seed-level-null split measured

Table 1 — The order, itemized and answered: the two links named at the close of Task 14's ledger, the theorem package or experiment that discharges each, its status, and its verification in this session's batteries

The standing honesty discipline of the corpus applies throughout: every number below is either computed exactly in this session's batteries (rank_defect.py, exp1_spectral.py, exp2_coarsening.py, exp3_sheaf.py, all with results JSONs) or it is labelled with its estimation error. Two mid-session retractions and repairs happened during the battery construction itself — an eigenvector biorthogonality bug that briefly misclassified the driven ring, and a truncation-convention transpose that briefly produced unstable reduced models — and both were caught by the anchor discipline before any claim was read off. The machinery sections below state the certificates that were checked.

2 · The objects and the classification

Fix the objects once. For a stationary process X , the arrow $A(X)$ is the time-reversal divergence: for a finite Markov chain with stationary distribution π and transition P , the exact rate $D = \sum_{i,j} \pi_i P_{ij} \ln(\pi_i P_{ij} / \pi_j P_{ji})$; for a process with a finite alphabet, the exact block reversal KL $\sum_u p(u) \ln(p(u)/p(\text{rev } u))$, computable by finite sums whenever the law is finite. The rank $\rho(X)$ is the rank of the covariance Hankel $[c(i+j)]$ of the observed process — the number of exponential modes of the autocovariance, the Kronecker-type law that Volume II's bridge theorems and Volume VII's analytic machinery have been orbiting all along. The complex count ρ_C is the number of visible modes with nonzero imaginary part. Task 14's discovery on the 3-ring was $\text{sv}_2 > 0$ iff driven; this volume asks what that equivalence really is.

The answer is a classification with four quadrants and four theorems. T1 (soundness): a visible complex mode is a sound arrow witness, because reversible chains are π -self-adjoint and therefore real-spectrum. T2 (the two failure modes): off the ring the equivalence fails in both directions — the hidden arrow, where an irreversible chain has an all-real spectrum (and the Birkhoff-polytope scan shows this quadrant has positive measure, 68.6 percent under the hit-and-run sampling), and the fake arrow, where a reversible process has complex covariance modes (the Gaussian AR(2), and the symmetric chain whose indicator sees two distinct real modes so that sv_2 is positive with zero arrow). T3 (the ring, every N): the equivalence's natural home — on the N-state cyclic ring the reversible slice is exactly the spectral-merging locus, the visible rank doubles when the arrow turns on, and the iff holds for every observable except the parity-affine family. T4 and T4b (the one-way valve and the reflection theorem): sensors are coordinate-wise channels, so the observed arrow never exceeds the source arrow and a reversible source stays reversible under every sensor; and on the ring every reflection is a reversal symmetry, so every reflection-invariant sensor launders the arrow exactly.

The witness	Arrow D	Hankel rank	Complex modes	sv_2 -witness	complex-witness
Driven 3-ring ($a=.32$, $b=.18$), identity observable	0.0596	2	2 ($0.25 \pm 0.1212i$)	consistent (the iff domain)	SOUND: complex $\Rightarrow$ driven
Doubly-stochastic [[.6,.4,0],[.2,.2,.6],[.2,.4,.4]]	$(1/15)\ln 3 = 0.0732$	1	0 (spectrum {1, 2/5, -1/5} exact)	HIDDEN ARROW (miss)	HIDDEN ARROW (miss)
Symmetric chain [[.5,.3,.2],[.3,.2,.5],[.2,.5,.3]]	0 (detailed balance exact)	2 (modes $\pm\sqrt{7}/10$)	0	FAKE ARROW (false positive)	silent (correctly)
Gaussian AR(2), roots $0.8 e^{\pm i\pi/5}$	0 (Toeplitz swap, error 0.0)	2	2	FAKE ARROW (false positive)	FAKE ARROW (the linear stratum launders)
Laplace MA(1), $x = \text{eps} - 0.7 \text{ eps}'$	0.0182 ± 0.0004 (exact integral MC)	2 (finite support)	0	HIDDEN ARROW (miss)	HIDDEN ARROW (miss)
Gaussian AR(1) (the OU baseline, process level)	0	1	0	silent	silent

Table 2 — The quadrant classification, witness by witness: the arrow D (the exact reversal divergence), the Hankel rank, the visible complex modes, and the verdict for both arrow witnesses (the Task-14 rank witness sv_2 and the complex-mode witness)

The last row of Table 2 is the OU baseline of Task 14 returned to its process-level home: reversible, single-mode, silent. The first row is the ring in the domain where the equivalence holds; the middle four rows are the boundary cases that break it in both directions. The doubly-stochastic witness deserves its own remark: its spectrum is certified in exact rational arithmetic — the characteristic polynomial coefficients match $6/5$, $3/25$, and $-2/25$ as fractions — and its arrow is exactly $(1/15) \ln 3$, while its indicator's covariance is a single exponential $(2/9)(2/5)^m$, verified term by term in fractions. The record of this chain is rank 1: the arrow is completely invisible to both witnesses, not because the chain is nearly

reversible but because its irreversibility is carried by the flow structure rather than the spectrum.

3 · Theorem T1 — soundness

The theorem: if an ergodic finite chain is reversible with respect to π_i , then P is self-adjoint in the π_i -weighted inner product, hence has real spectrum; therefore a visible complex eigenvalue is a sound witness of non-reversibility, and the visible complex mode of the observed process's covariance is a sound witness of the arrow. The proof is two lines — detailed balance $\pi_i P_{ij} = \pi_j P_{ji}$ says exactly that the π_i -symmetrized matrix is symmetric — but its contrapositive is the engine of every arrow witness this programme has used: complex spectrum implies irreversible implies $D > 0$. What the theorem does not say, and what the next section witnesses, is the converse.

The verification is exhaustive on the sampled families. One thousand random walks on one thousand random undirected graphs (N between 3 and 8) are reversible by construction (degree stationary, detailed balance verified to a maximum violation of $8.3e-17$), and every one has an all-real spectrum — the maximum imaginary part over the entire sample is $6.8e-17$, machine zero. In the other direction, every sampled chain with a complex eigenvalue has strictly positive reversal KL, four thousand out of four thousand. The soundness direction is not a probabilistic statement; it is a theorem verified to exhaustion because it could not fail. The interesting mathematics lives in the two ways the converse fails, and in the exact structure that makes the ring the place where it does not fail at all.

4 · Theorem T2 — the two failure modes

The hidden arrow first. The witness chain of Table 2 is doubly stochastic (so the stationary distribution is uniform and the reversal KL has the clean closed form $(1/15) \ln 3$), non-symmetric, and carries the rational spectrum $\{1, 2/5, -1/5\}$. Its indicator observable sees exactly one exponential mode, so the Hankel rank is 1 and sv_2 is zero at machine precision: both the Task-14 rank witness and the refined complex-mode witness report nothing while the arrow is strictly positive. At the process level the Laplace MA(1) does the same with continuous state: its autocovariance has finite support (rank 2, all modes degenerate at zero), yet its 2-block reversal KL is 0.0182 with a standard error of 0.0004, computed by an inner integral that is exact — the log-integrand is piecewise linear in the shared innovation, so each segment integrates in closed form — and only the outer expectation is Monte Carlo. The same machinery with Gaussian innovations returns exactly zero, the classical boundary: the MA(1) is reversible if and only if its innovations are Gaussian.

The fake arrow second. The symmetric chain is reversible by construction with distinct real eigenvalues $\{1, +\sqrt{7}/10, -\sqrt{7}/10\}$, so its indicator sees two distinct exponential modes and sv_2 is strictly positive with zero arrow: the rank count is a mode count, not an arrow detector. The Gaussian AR(2) with

roots $0.8 \exp(\pm i \pi/5)$ is sharper still: its covariance Hankel has rank 2 with a genuine complex-conjugate mode pair, and its reversal KL is exactly zero — the block covariance is Toeplitz in $|i-j|$ and therefore swap-invariant, verified to 0.0 at block lengths 2, 4, and 8. Every real stationary Gaussian process is reversible; the linear-Gaussian stratum launders the complex-mode witness exactly the way the reflection sensors launder the ring's arrow in Theorem T4b. Soundness of the complex witness is a jump-process theorem; the continuous linear stratum is exempt.

And the failure is generic, not exotic. The hit-and-run sampler on the Birkhoff polytope B_3 — four-dimensional, uniform-stationary chains — finds that 68.6 percent of sampled doubly-stochastic 3-state chains have an all-real spectrum with strictly positive reversal divergence, with the divergence ranging up to 1.39. For random 4-state stochastic matrices the all-real fraction is 36.8 percent. The hidden-arrow quadrant has positive measure: the equivalence arrow iff rank defect fails generically off the abelian slice, in exactly the direction the doubly-stochastic witness exemplifies. The ring was never the typical case; it was the structured case, and the next theorem says precisely what the structure is.

5 · Theorem T3 — the ring, every N

On the N -state cyclic ring with forward rate a , backward rate b , and self-loop $1-a-b$, the transition matrix is circulant, so its spectrum is the closed form $\lambda_j = 1-a-b+a \omega^j + b \omega^{-j}$ with $\omega = \exp(2 \pi i / N)$. The imaginary part is exactly $(a-b) \sin(2 \pi j / N)$ — verified machine-exact for every N from 3 to 8 — so the spectrum is real if and only if $a = b$, and at $a = b$ the spectrum is real and degenerate: the pairs $(j, N-j)$ merge. This is the mechanism the 3-ring exhibited and the N -generalization makes explicit: the reversible slice is the spectral-merging locus. The observable's covariance sees one exponential per distinct eigenvalue, so the visible rank at equilibrium is $\text{floor}((N-1)/2)$ plus one more if N is even, and the driven rank is $N-1$ — every pair splits into its conjugate halves. Table 3 records the law for $N = 3$ to 8, six of six, with the defect δ equal to $\text{floor}((N-1)/2)$.

N	Equilibrium rank	Driven rank	Defect δ	Predicted δ	Circulant spectrum error
3	1	2	1	1	machine-exact
4	2	3	1	1	machine-exact
5	2	4	2	2	machine-exact
6	3	5	2	2	machine-exact
7	3	6	3	3	machine-exact
8	4	7	3	3	machine-exact

Table 3 — The N -ring rank-doubling law, $N = 3..8$: the equilibrium rank (the pair-merging), the driven rank, the defect, and the closed-form check $\text{Im } \lambda_j = (a-b) \sin(2 \pi j / N)$ — the reversible slice IS the spectral-merging locus, which is exactly why the ring satisfies the iff

The boundary is as sharp as the law. The equivalence holds on the ring for every observable whose DFT support touches a pair $(j, N-j)$ with j not 0 or $N/2$ — for odd N that is every non-constant observable. For even N the parity-affine observables, the functions of $x \bmod 2$, see only the real $j = N/2$ mode: their rank profile is the same at equilibrium and under driving, and the arrow is invisible to them. The parity sensor on the driven 4-ring realizes this exactly — the observed process is the flip/stay chain, reversible, block reversal KL zero at every block length to $1.7e-18$ — which is not a degenerate truncation but the sharpest statement of the whole theorem: memory without the arrow, a record that keeps its correlation and loses its handedness.

6 · Theorems T4 and T4b — the one-way valve and the reflection laundering

The one-way valve is the data-processing inequality in the record's clothing. A sensor is a coordinate-wise channel applied to the source process; the block law of the observed record is the pushforward of the source's block law; and the KL divergence contracts under pushforward. Two consequences: the observed arrow never exceeds the source arrow (verified in exact block arithmetic at every block length on every sensor of the laundering lattices), and a reversible source stays reversible under every sensor — the lumped symmetric chain's block KL is zero at every block length, to $5.7e-18$. The record cannot fabricate the arrow. It can only inherit it, degrade it, or destroy it, and the destruction has an exact characterization that the battery discovered in its own zeros.

The exact zeros came first as a surprise: on the 3-ring, every single non-identity sensor launders the arrow completely — the block KL is zero at every block length for $\{0\}\{1,2\}$, for $\{0,2\}\{1\}$, for $\{0,1\}\{2\}$, and trivially for the constant sensor. The theorem behind the zeros is the reflection-laundering theorem: if a symmetry ρ of the state space conjugates the dynamics to its own time reversal ($\rho P \rho^{-1}$ = the reversed chain), then every ρ -invariant sensor launders the arrow exactly — the reversal of the law is the ρ -conjugate of the law, and the sensor is ρ -blind. On the N -ring every reflection is such a reversal symmetry, because reflecting the ring swaps clockwise and counterclockwise, which is exactly swapping a and b , which is exactly the stationary time reversal at uniform π . So the ring's arrow is its chirality — its circulation direction — and the record loses the arrow exactly when it loses the handedness.

The theorem's predictions were then tested against a designed counterexample: on the driven 4-ring, the three-block sensor $\{0\}\{1\}\{2,3\}$ breaks all four reflections, so it is chirality-keeping and the arrow must survive — and it does, with a strictly positive block KL of 0.453 at block length 7, while both reflection-orbit controls ($\{0\}\{1,2,3\}$ and the two-block sensors) launder to exactly zero. Figure 1(b) shows the split: the identity and chirality-keeping sensors hold the arrow at every block length, growing linearly at the reversal rate, while the parity and half sensors sit on the axis at machine zero. The 3-ring, having no chirality-preserving coarsening other than the identity, is the minimal chiral record: its arrow survives

only at full resolution. The complex pair of the rank-doubling law is the chiral mode; killing the handedness kills both.

7 · The scope theorem — the answer

The classification assembles into the scope theorem, the answer to the ordered question. The processes satisfying arrow iff rank defect are exactly those whose non-reversibility is spectrally carried inside the observable's visible spectrum. The circulant (abelian) families with arrow-carrying observables are the canonical class: within them the reversible slice coincides with the spectral-merging locus, so turning the arrow off forces the degeneracy that collapses the rank, and the equivalence is a structural identity rather than a coincidence. Beyond the abelian class both failure quadrants have positive measure — the real-spectrum irreversibles of the hidden arrow and the symmetric mode-splittings and Gaussian complexes of the fake arrow — and the equivalence becomes the exception that the ring's structure buys.

In the corpus's language: the rank defect is the abelian shadow of the arrow. The arrow itself — the reversal divergence of the full law — is the record-theoretic object, monotone under sensors, inherited or destroyed but never fabricated; the Hankel rank sees only the second-moment, abelianized projection of that law, and on the abelian families the projection is faithful. This is the same abelianization that Volume VIII measured algebraically (the Parikh reduction, the circulant class where the transport sandwich closed): the rung was the place where the free structure collapsed to its commutative shadow, and it is the same place where the arrow becomes a rank. The BT2 stratum (the Hankel memory) and the BT3 stratum (the reversal intercept) meet at the register, and the meeting is exact only on the abelian slice.

The chat's final synthesis lands here with its correction intact: 'it and bit from record' requires the record to have a history, and the one-way valve says the history's asymmetry is bounded by the world's. A record that launders the chirality — the parity sensor, the reflection orbits, the Gaussian stratum — can carry complete information about correlation structure while carrying none of the arrow; a record that keeps the chirality keeps an arrow it cannot amplify. The three strata of Task 14's stratification (classical reality, thermodynamic irreversibility, quantum coherence) now have their sensor-theoretic boundary: the arrow is a chirality carried by the circulation, the record's fidelity to it is measured by the block reversal KL, and the rank defect is the abelian shadow that fidelity casts on the covariance.

8 · Experiment 1 — the setup and the compression

The chat's first experiment names WikiText-103 or 'a synthetic task with known finite-state structure' as its benchmarks; the second branch is the scientifically cleaner one and the one this battery takes. The true process is this volume's own object: the driven 6-ring ($a = 0.32$, $b = 0.18$) observed through the 4-symbol sensor $s(i) = i \bmod 4$, an HMM whose observed process has a known minimal realization. The exact joint-probability Hankel over 64×64 (prefix, suffix) contexts of length 3 returns machine-exact rank 5 — the sensor's collisions (states 0 and 4 emit the same symbol, 1 and 5 likewise) cost the process one dimension against its 6 hidden states — and the rank is stable from window 3 to window 4. The entropy rate, computed by the exact forward filter over four hundred thousand steps, is 0.9559 nats per symbol with a Monte Carlo standard error of a few parts in a thousand. The models: a linear state-space RNN ($h' = hA + xB$, logits $= hC + c$, $d = 16$) trained by backpropagation through time, whose Hankel singular values and reductions are exact closed-form objects via the Gramian machinery; and a vanilla tanh RNN ($d = 16$), the classic recurrent model, measured empirically. Three seeds each, identical regime.

The compression claim is confirmed with an honest refinement. The tanh RNN reaches a cross-entropy of 0.959 against the entropy floor of 0.9559 — essentially optimal prediction — from a state whose spectrum collapses from 16 dimensions to 3 at the one percent tolerance (3.0 across seeds; 2.7 at five percent). The linear RNN reaches 0.979 with a Gramian profile whose effective rank crosses the true rank 5 at the five percent tolerance (7.3 at one percent, 6.0 at two, 5.0 at five, 4.0 at ten). The learned transition spectra carry the chiral complex pair in all three linear seeds — the models learn the record, not a smoothed surrogate; they rediscover the arrow's spectral signature. But the task knee is not at the WFA rank: the cross-entropy curve flattens at order 3, the predictive (sufficient-statistic) dimension, matching the tanh model's 3-dimensional state and the sigma profile's own elbow ($\sigma_2 = 4.77$ dropping to $\sigma_3 = 0.89$). The linear realization rank overcounts the information dimension by the modes that carry the joint law but not predictive entropy.

The machinery was certified before it was trusted, in the Vol IX discipline: the Gramian fixed-point solution agrees with the direct Lyapunov series; the balanced-transform certificate (the balanced controllability Gramian equals diag of the squared sigma) holds at $8e-16$; and the exact error-system Hankel norms — computed from the composite system's Gramians, no truncation, no tail bounds — agree with a direct 30×30 -block Hankel SVD to four digits on a random test system, with the boundary-order truncation hitting the AAK floor exactly. One mid-session transpose bug in the system convention (the model's row-vector dynamics against the column-vector Lyapunov equations) produced unstable reduced models and was caught by exactly these certificates; the numbers below are from the repaired, re-verified machinery.

9 · Experiment 1 — the AAK extraction and the baselines

The extraction is the square-root balanced truncation at orders 1, 2, 3, 4, 5, 6, 8, 10, 12, and 16, with the exact Hankel-norm error of every truncation computed from the error system. The AAK interval holds ten of ten: the measured error lies between the floor σ_{n+1} and the tail sum at every order, as the theorem demands — the optimal rank- n Hankel approximant cannot beat σ_{n+1} and the balanced truncation cannot beat the tail. The chat's prediction that 'the distortion-rate curve will match the predicted σ_{n+1} lower bounds' is thus approximately true in the strongest honest sense available: the measured ratios of error to floor are 1.00 at the boundary orders $n = 1$ and 2, 1.00 again at the last surviving mode, and between 1.02 and 1.31 inside — the truncation is AAK-optimal exactly where the tail is negligible, and the profile shows both regimes. The exact curve-match is refuted, as different norms must disagree; the knee prediction is exact.

The baseline adjudication splits honestly. Against random state deletion the AAK extraction wins at every order — 1.11 to 1.48 against 0.98 to 1.38 on the cross-entropy scale — and against magnitude deletion it wins or ties essentially everywhere. But against direct retraining of an order- n model, the truncated big model loses at the lowest orders: at $n = 1$ the fresh small model reaches 1.12 against 1.30, at $n = 2$ it reaches 1.04 against 1.38, and from $n = 3$ they tie at the parent model's level. The chat's claim that the AAK extraction 'outperforms existing distillation baselines at the same parameter count' is confirmed against the pruning baselines and refuted against matched-size direct retraining at the bottom orders: the truncated model inherits structure a tiny model cannot use, and retraining adapts. Figure 1(c) records the split. The deeper reading is the corpus's own: the sigma spectrum is a compression budget — the elbow at σ_3 predicts the task knee exactly — but the budget counts the linear realization's modes, and the sufficient statistic is smaller than the realization.

10 · Experiment 2 — the sensor-coarsening scan

The chat's second experiment demands a POMDP benchmark with a restricted sensor; it names RockSample, Hallway, and 'a sensor-limited version of the Tiger problem,' and this battery takes the Tiger branch because it admits exact machinery at every coarsening. The canonical Tiger: hidden side, open-left or open-right for +10 correct or -100 wrong, listen at a cost of 1 with accuracy κ . The listen channel is the sensor; coarsening it from 1.0 down to 0.5 is the scan. The solver is the finite-horizon ($h = 8$) alpha-vector backup with exact pruning — the upper-envelope monotone chain drops only never-maximal lines, so the value is exact at every belief — and the pruning itself returns the policy's belief regions and their kinks in linear time. The obstruction datum is the chat's 'finite-state policy fails' made exact: the gap between the optimal value and the best memoryless one-state controller, computed by enumerating all twenty-seven controllers and solving each closed chain. The register is the alpha-count. The curvature is the interior kink mass of the piecewise-linear value function, weighted and unweighted by the exact closed-loop belief visitation.

The register law is the scan's headline. The optimal policy's alpha-count grows from 9 at $\kappa = 1.0$ through 59 at 0.95, 34 at 0.90, 44 at 0.80, 45 at 0.70 — and then explodes to 9153 at $\kappa = 0.60$: the curse of history made visible in a single number. The poorer the sensor, the more belief regions the optimal controller must distinguish, until the game itself collapses to the stall regime (at $\kappa = 0.55$ and 0.5 the optimal value is listen-out-the-horizon at -8, and the register collapses back to 280 and 4). This is Volume VIII's register inflation — there measured algebraically by the box-determinant law on the abelianized rung — measured here on the control side, exactly, on the chat's own benchmark. The obstruction iff holds at every κ : $o(\kappa) > 0$ exactly when the optimal policy needs memory, including the stall boundary where the optimum itself is memoryless and $o = 0$. The chat's weak form is verified at both ends: at $\kappa = 1$ the obstruction vanishes and the one-state controller is optimal.

The quantitative predictions fare worse, and the volume records the numbers rather than rounding them up. On the informative range (κ at or above 0.7, where the game is not stall-dominated) the obstruction correlates with the oracle gap at Pearson $r = 0.38$ and the register at 0.39 — weak; the chat's 'curvature predicts the magnitude of the gap' is refuted in its weighted form outright: the visit-weighted curvature anti-correlates at $r = -0.95$, because as the gap grows the closed-loop visitation concentrates on the flat listen-and-stall paths where the value function has no kinks. The raw interior kink mass saturates near its telescoping ceiling. What survives is the structure: the iff exact, the register inflation dramatic, and the finite-horizon stall identified as the honest confound — the memoryless controller's best coarse option is to avoid the catastrophe by never committing, which caps its loss and decouples the obstruction from the gap.

The record side of the experiment returned a theorem instead of a curve, and it is worth stating as such: the Tiger's listen record is conditionally iid given the hidden side, so every block law is the exchangeable

mixture one-half product of $P(u_t \text{ given } L)$ plus one-half product of $P(u_t \text{ given } R)$, which is invariant under index reversal — the arrow of the record is identically zero at every sensor quality, verified on all sixteen 4-blocks at every kappa to a maximum deviation of $3.5e-18$. The chat's bridge from control performance to the arrow of time cannot be tested on memoryless-sensor benchmarks: the record stratum degenerates by construction. The bridge lives on sources with temporal structure — exactly the class Part 1's laundering theorems cover — and the experiment, honestly run, separates the strata it was meant to conflate.

11 · Experiment 3 — the sheaf diagnostics

The chat's third experiment constructs a cellular sheaf from a network's local computations and predicts that nonvanishing cohomology tracks training failures, with the harmonic norm correlating with the validation gap; its named benchmarks are SCAN and COGS for compositional generalization. The battery takes the compositional branch at its cleanest: modular addition on $\mathbb{Z}_7$, (a, b) mapping to $(a+b) \bmod 7$, the minimal two-token-family compositional task, with a 2-64-64-7 ReLU MLP trained to convergence (twelve seeds per split) on four splits — the compositional quadrant split (train a and b in $0..3$, sixteen pairs, thirty-three held out), an intermediate split (thirty-six and thirteen), the full-grid control, and a scattered random-60 split. The sheaf is the cellular sheaf on the bipartite token graph: fourteen token vertices (seven row, seven column), the training pairs as edges, vertex stalks the hidden activation space, and the edge restriction maps the per-token PCA projectors of that token's training activations. The diagnostic is the coboundary energy of the identity section: for each training edge, the normalized disagreement between the row-token and column-token reconstructions of the shared example's activation — the failure of the local representations to glue into a global section, the honest computable instantiation of the chat's harmonic norm.

The compositional failure reproduces exactly as the literature reports it: training accuracy 1.000 on every split and every seed, while held-out accuracy is 0.073 on the compositional split, 0.006 on the intermediate, 0.000 on the scattered random-60, and 1.000 on the full control. The network memorizes its training pairs and fails to glue the token geometries — including the scattered split's held-out compositions of seen tokens, where both tokens have trained subspaces and the composition is the only new thing. The sheaf's verdict splits cleanly between task-level and seed-level, and the split is the experiment's honest result. Task-level: the mean coboundary energy orders the four splits exactly as their OOD errors do — 0.541 for the full control against zero OOD error, 0.577 for the compositional against 0.927, 0.594 for the intermediate against 0.994, 0.615 for the random-60 against 1.000 — a perfect ordering. The energy predicts which task geometry fails.

Seed-level: within a split, the correlation between the energy and the OOD error across the twelve seeds is null to slightly negative (-0.20 and -0.18 on the two informative splits, 0.0 where there is no variance to correlate). The pooled correlation of 0.59 is dominated by the split separation, and the volume flags it as

such: the sheaf diagnostic is a geometry-of-the-task predictor, not a which-random-seed-fails predictor. The chat's claim 'the harmonic norm correlates with the validation gap' is therefore confirmed at the task level and refuted at the seed level on this benchmark — the honest split that SCAN-scale experiments with single runs cannot even express, because they cannot separate the two variances. The edge-level diagnostic on the random-60's both-trained held-out pairs completes the picture: the token-reconstruction conflict on unseen compositions, measured per run, is the sheaf's local instrument, and its correlation with the per-run error is recorded in the results JSON at 0.0 — the local instrument sees the geometry, not the seed.

12 · The ledger

What survived this session: the four theorems of the classification (soundness verified to exhaustion, the rank-doubling law six of six with the closed-form mechanism, the one-way valve exact, and the reflection-laundering theorem discovered, proved, and tested against a designed counterexample); the scope-theorem answer to the ordered question — the equivalence holds exactly where the non-reversibility is spectrally carried, the abelian families with arrow-carrying observables, and the rank defect is the abelian shadow of the arrow; and the three experiments run at full audit strength with their predictions split honestly — the AAK interval exact and the knee exact, the register explosion measured, the sheaf ordering perfect at the task level. What was refuted: the chat's distortion-rate curve-match (approximately true, exactly false, ratios 1.00 to 1.31), its AAK-beats-retraining claim (reversed at the lowest orders), its curvature-predicts-the-gap claim (anti-correlated in the weighted form, confounded by the stall), and the seed-level form of the sheaf-correlation claim. What was discovered: the reflection-laundering theorem and the chirality reading of the ring's arrow; the exchangeability degeneracy of memoryless-sensor records; the predictive-dimension-versus-realization-rank gap in sequence-model compression; and the positive measure of the hidden-arrow quadrant.

The corpus integration lands on three prior volumes. Volume VIII's abelianized rung is the algebraic home of the scope theorem: the circulant class where the transport sandwich closed is the same class where the arrow is a rank, and the register inflation it measured by the box-determinant law reappears here as the Tiger's alpha-count explosion under sensor coarsening — the same law measured on both sides of the observation boundary. Volume IX's exact-norm discipline is the machinery that made the AAK interval and the balanced-truncation certificates trustworthy, and the mid-session transpose bug that the certificates caught is the discipline working exactly as designed. Task 14's stratification record completes: the three strata are now separated by theorems rather than by slogans — the memory stratum (the Hankel rank, the BT2 side), the arrow stratum (the reversal divergence, the BT3 side, a chirality that sensors can launder exactly), and their meeting at the register, exact only on the abelian slice. 'It and bit from record' now has its sensor-theoretic content: the record is the pushforward, its arrow is bounded by the world's, and its rank is the abelian shadow the world casts when the world is abelian enough.

The remaining open links are unchanged from Volume IX's ledger and are carried forward honestly: the cell's exact $D(2)$ between the measured walls $[1.000, 1.2771]$, with the rank-wall obstruction standing; Open 7.13, the non-commutative AAK construction, which the abelian-shadow theorem of this volume now frames precisely — the free-monoid approximants live beyond the abelian slice, exactly where the equivalence stops being structural; Φ 's uniqueness; Risk 4; the bounded benchmarks; and the $n = 4, L = 10$ leg of the record-phase scan. The next natural link this session's mathematics opens: the reflection-laundering theorem for non-ring dynamics — which irreducible chains admit a reversal symmetry at all, and what the lattice of arrow-preserving sensors looks like off the circulant class — is stated as an open problem with the 3-ring's minimality (the identity is its only handedness-preserving sensor) as the first datum.

4 + 1 theorems proved or discovered this session (T1-T4, the reflection-laundering theorem)	68.6% of B3 has a real spectrum with $D > 0$ — the hidden-arrow quadrant's positive measure	9 → 9153 the optimal Tiger controller's register explosion as the sensor coarsens to 0.6
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